

COMMUNICATIONS RECEIVER

FRG-7



GENERAL DESCRIPTION

The model FRG-7 is an all solid state synthesized communication receiver designed to cover the entire high frequency spectrum, 500 kHz to 29.9 MHz.

FRG-7 is a triple conversion super heterodyne receiver utilizing synthesized heterodyne oscillator known as the "Wadley Loop System" which offers unparalleled stable performance.

The calibrated dial mechanism provides 10 kHz frequency readout throughout the receiver coverage.

Good selectivity is provided for SSB, AM and CW with the utilization of a ceramic filter in the 455 kHz IF circuits.

The FRG-7 includes three step front end attenuator, amplified AGC and low-normal-high tone select switch for extreme flexibility that even the most demanding amateur, CBER, or broadcast listener desires. In addition, the large cabinet and hi-fi speaker will provide you with high quality audio output.

The FRG-7 includes a self-contained three way power supply for 100/110/117/200/220/234 volts AC 50/60 Hz, an internal battery or external 12 volt DC. If the AC power source fails, the unit switches automatically to an internal battery which uses eight UM-1 dry cells.

To save battery consumption, the dial lamps can be switched off

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SPECIFICATIONS

Frequency Range:

0.5 MHz 29.9 MHz

Type of Emission:

AM, SSB (USB or LSB), CW

Sensitivity:

SSB/CW: Better than 0.7 MV at S/N 10 dB

AM : Better than 2 MV at S/N 10 dB

Selectivity: ± 3 kHz at -6 dB, ± 7 kHz at -50 dB**Stability:**Less than ± 500 Hz at any 30 minutes after warm up**Antenna Impedance:**High impedance for 0.5 MHz ~ 1.6 MHz 50 ohm
unbalanced for 1.6 MHz ~ 29.9 MHz**Speaker Impedance:**

4 ohms

Audio Output:

2 watts

Power Requirement:100/110/117/200/220/234 volts AC 50/60 Hz, 12 volts DC
external or internal dry cell UM-1x8**Power Consumption:**

AC 14VA

Size:

340(W), 153(H), 285 (D)mm

Weight:

Approx. 7 kg without batteries

SEMICONDUCTORS COMPLEMENTS

IC:

AN-214	1	SN76514	1
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FET:

3SK-40	3	2SK19	6
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Transistor:

2SC372	8	2SC784	4
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2SD313	1		
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Diode:

1N60AM	9	1S1555	2
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VO6B	3		
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Zener Diode:

WZ-1 10	1	BZO9I	1
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INSTALLATION

Carefully remove the FRG-7 receiver from the carton and examine it for any physical damage.

Should any be apparent immediately notify the carrier stating the damage in detail. Save the carton and packing materials for future use.

Location:

In general, the location of the FRG-7 is not critical, however, it is recommended that excessively warm location be avoided.

POWER REQUIREMENT

The FRG-7 is supplied with a multi-voltage power transformer (export model only) and can be operated in many areas of the world where supply voltage may differ from your local supply voltage. Therefore before connecting the AC cord to the power outlet, be sure that the voltage marked on the rear of the receiver agrees with the local AC supply voltage.

CAUTION

PERMANENT DAMAGE WILL RESULT IF IMPROPER AC SUPPLY VOLTAGE IS APPLIED TO THE RECEIVER.

The FRG-7 will operate satisfactorily from any 12 volt, negative ground battery source by connecting the DC power cord (plug is supplied) to the rear panel receptacle. When making connections to the battery, be certain that the inner conductor is connected to the positive (+) and the outer conductor is connected to the negative (-) terminals of the battery. Reversed connection could permanently damage the receiver circuit.

The FRG-7 will also operate from eight dry cells in the built-in dry cell pack. (Cells are not supplied) If the AC supply fails, the dry cell supply is automatically connected to the circuit.

The following Table I shows the power supply combination of FRG-7.

Power Source	1	2	3	4	5	6	7
AC Supply	O	-	-	O	O	O	-
External DC	-	O	-	X	-	X	O
Internal DC	-	-	O	-	X	X	X

O Power source in use

X Power source connected but not in use

- Power source not connected

Table 1

ANTENNA AND GROUND

The antenna is the most important part of the communication receiver installation. The FRG-7 is designed for use with a long wire antenna for 0.5 ~ 1.6 MHz and with a resonant antenna at the operating frequency having an impedance of 50 to 75 ohms for higher frequency than 1.6 MHz. This requirement is easily met by using a center fed dipole antenna resonated to the receiving frequency and fed with coaxial cable.

The FRG-7 should be connected to a good ground. The ground lead should be connected to the terminal marked E located on the rear panel of the receiver.

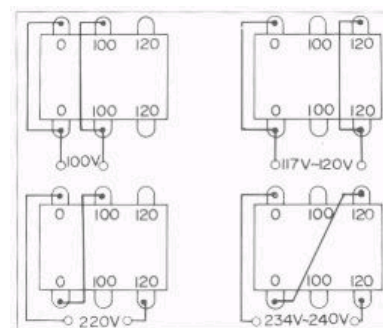
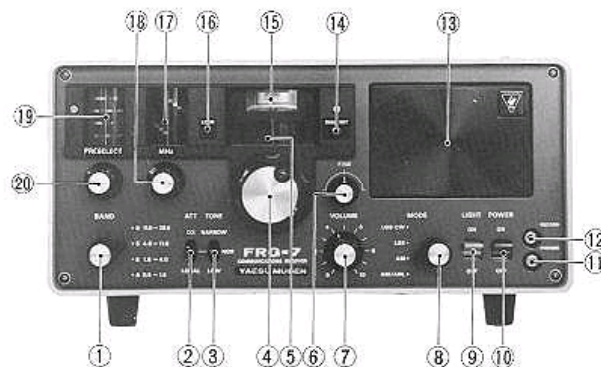


Fig. 1 Transformer Primary Wiring

CONTROLS AND SWITCHES

The FRG-7 has been designed for ease of operation. All controls have been properly adjusted at the factory. Several panel controls and switches are unusual in operation, and an improper setting may result in poor reception. The function of various controls and switches is described in the following paragraph.

Be certain that you thoroughly understand the individual function of each before operating the receiver.



FRONT PANEL

(1) BAND

The BAND switch is a four position switch. The switch selects the desired frequency range.

(2) ATI (NOR. DX, LOCAL)

The ATT (attenuator) switch attenuates the incoming signal to prevent over-loading of the front end when an extremely strong signal is present. At the switch NOR (normal) position, the attenuator is removed from the input circuit.

(3) TONE (NOR, NARROW, LOW)

The TONE switch changes audio response of the receiver. The audio amplifier passes at the NOR position, 250 Hz through 3000 Hz, at NARROW 400 Hz through 2500 Hz band at LOW 250 Hz through 1500Hz.

(4) (5) TUNING DIAL

The main TUNING knob determines the frequency

in combination with the BAND switch and MHz setting.

(6) FINE TUNING

The FINE TUNING control is used for precise tuning of the received signal. The main tuning dial is calibrated to the frequency with the fine control at centre.

(7) VOLUME

The VOLUME controls the audio output level from the speaker.

(8) MODE

The MODE switch determines the appropriate detector in use. In the USBCW position, the USB (Upper Side Band) and code signal is heard. In the LSB position, the LSB (Lower Side Band) signal is heard. In the AM position, the amplitude modulated signal is heard and the Noise Limiter is put into the circuit in the AM/ANL position.

CONTROLS AND SWITCHES (continued)

(9) LIGHT

This switch is used to turn off the lamp so as to save the current drain when the FRG-7 is operated from internal dry cells.

(10) POWER

This switch turns off the supply voltage for 1)0th AC anti DC operation.

(11) PHONES

Phone jack is provided for private listening and the speaker is disconnected when the plug is in-serted in this jack.

(12) RECORD

This jack is for recording purpose and the output level is set to approximately 50 mV regardless of setting of the VOLUME control.

(13) SPEAKER

Internal Speaker.

(14) DIAL SET

Main tuning dial calibrator.

(15) S-METER

The S-meter indicates the relative signal strength of the received signal. It is calibrated in S-unit from S-1 to S-9 and in dB over S-9.

(16) LOCK

The LOCK lamp lights up when the synthesized heterodyne oscillator is unlocked.

(17) (18) MHz

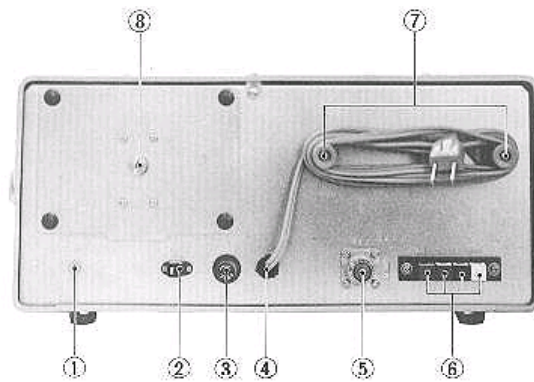
This MI-lz control synthesizes heterodyne oscillator to the harmonics of 1 MHz crystal oscillator. The scale is calibrated in MHz with the frequency show-ing the correct setting of the heterodyne signal.

(19) (20) PRESELECT

The PRESELECTOR control tunes the receiver front end. The scale is calibrated with the fre- ~ quency showing the correct setting for various bands.



Fig. 2 Headphone Connection



Rear Panel Connection

(1) EXTSP

This jack is for connection of a 4 ohm external speaker when desired. With the plug in the S jack, the internal speaker is disconnected.

(2) EXT DC

Receptacle for external 12 volts DC supply.

(3) FUSE

Fuse for AC operation. Use 0.15 amp rating fuse.

(4) AC cord

Cord for AC operation.

(5) SW2

Coaxial connector for short wave listening.

(6) SW,BC,E,MUTE

SW is long wire antenna terminal for the short wave listening.

BC is long wire antenna terminal for the broadcast band listening.

E is ground connection.

MUTE is used to disable the receiver while transmitting. Connect this terminal to ground for receiver muting.

(7) (8)

AC cord holder and the internal battery pack. Use eight UM-1 dry cells.

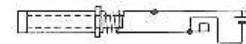


Fig.3 External Power Plug Connection

OPERATION

FREQUENCY SELECTION

The receiving frequency is selected by the combination of the MHz dial and main tuning dial settings. The MHz dial selects the band at every 1 MHz and the main tuning dial selects the frequency at 10 kHz increment from 0 to 990 kHz in the band. The combination of these controls is shown in Table 2.

AMATEUR BAND RECEPTION

SSB Voice Signal:

Most amateurs use LSB on frequencies lower than 10 MHz and USB on frequencies higher than 10 MHz.

Set the controls and switches as follows;

POWER OFF

BAND Desired frequency segment

ATT NOR

TONE NOR

VOLUME Desired listening level

	Frequency	PRESELECT	MHz	Main Dial	BAND	MODE
Amateur	1,910	2.0 1.9	1	910	B1.6-4.0	USB - CW
	3,595	3.6 3.3	4	595	B1.6-4.0	LSB
	7,050	7 4	7	050	C4.0-11.0	LSB
	14,175	15 13	14	175	D11.0-29.9	USB - CW
	21,225	23 20	21	225	D11.0-29.9	USB - CW
	26,850	30 26	26	850	D11.0-29.9	LSB - CW
Medium Wave	550	0.8	0	550	A0.5-1.6	AM or AM/ANL
	960	1.0 0	0	960	A0.5-1.6	AM or AM/ANL
	1,170	1.2 1.0	1	170	A0.5-1.6	AM or AM/ANL
WWW.DUT	2,300	2.6 2.3	2	300	B1.6-4.0	AM or AM/ANL
	5,000	5 4	5	000	C4.0-11.0	AM or AM/ANL
	10,000	11 10	10	000	C4.0-11.0	AM or AM/ANL
	15,000	17 15	15	000	D11.0-29.9	AM or AM/ANL
Short Wave	3,525	4.0 3.6	3	525	B1.6-4.0	AM or AM/ANL
	5,580	6 4	5	580	C4.0-11.0	AM or AM/ANL
	6,715	10 9	6	715	C4.0-11.0	AM or AM/ANL
	11,705	12 11	11	705	D11.0-29.9	AM or AM/ANL
	15,120	17 15	15	120	D11.0-29.9	AM or AM/ANL
	17,850	20 17	17	850	D11.0-29.9	AM or AM/ANL
	21,550	23 20	21	550	D11.0-29.9	AM or AM/ANL

Table 2

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MODE LSB for 160, 80 and 40 meter bands
 USB for 20, 15 and 10 meter bands

DIAL SET Center

PRESELECT Desired frequency. Refer to Table 2.

MHZ ""

MAIN DIAL ""

CW (Morse Code Signal):

The code signal can be heard with the MODE switch at USB/CW position and by tuning the main tuning dial for a desired listening tone.

Turn the LOWER switch on. Precisely adjust the MHz dial until the LOCK lamp turns off. Tune the main tuning dial for the desired signal until the signal is clearly heard. Use the FINE TUNING control for precise tuning. When the received signal is garbled, try the opposite sideband. When an extremely strong signal is distorted, peak the PRESELECTOR for a maximum S-meter reading.

BROADCAST RECEPTION

The broadcast signal is transmitted on AM mode. If impulse type noise is experienced, set the MODE switch to AM/ANL position to reduce the noise interference.

Set the ATT switch to LOCAL position to avoid front end over loading. Set the VOLUME for desired listening level.

The amateur SSB signals cut high and low audio response, so that it may be helpful to reduce the interference by setting the TONE switch at NARROW or LOW position.

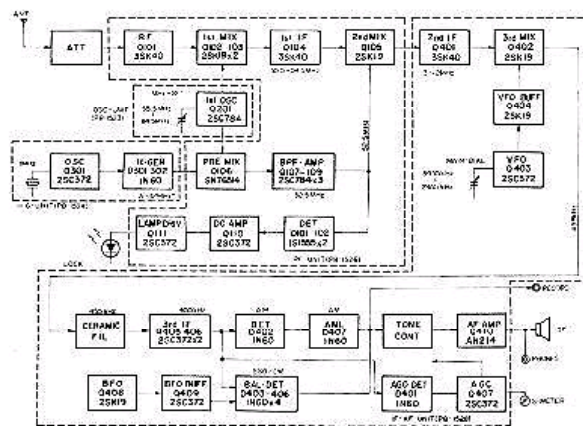


Fig.4 BLOCK DIAGRAM

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CIRCUIT DESCRIPTION

The block diagram will provide you with a better understanding of this receiver. In general, the FRG-7 is a tripple conversion super heterodyne receiver utilizing synthesized local oscillator for both the first and second mixers for drift free VFO operation.

The signal from the antenna is fed through the attenuator to the gate of the FET RF amplifier Q₁₀₁, **3SK40**. The amplified signal is fed through a low pass filter (cut off frequency 35 MHz) to the first balanced mixer consisting of Q₁₀₂ and Q₁₀₃, **2SK19**, where the incoming signal is mixed with a signal from the heterodyne oscillator. The first heterodyne oscillator Q₂₀₁, **2SC784**, oscillates the signal which varies between 55.5 and 84.5 MHz.

The product of the first mixer becomes the first IF signal of 54.5 through 55.5 MHz. The first IF signal is amplified by the first IF passband amplifier Q₁₀₄ and fed to the gate of the second mixer Q₁₀₅, **2SK19CR**, where the first IF signal is mixed with 52.5 MHz signal. The second mixer converts the first IF signal into the second IF signal of 2.0 through 3.0 MHz.

Synthesizer oscillator Q₃₀₁, **2SC372**, oscillates crystal controlled 1 MHz signal. The 1 MHz signal is then fed to the harmonic generator D₃₀₁ and D₃₀₂, **1N60**, which produces 3 to 32 MHz harmonics from the 1 MHz crystal controlled signal. The harmonic signal is fed to the dual balanced pre-mixer Q₁₀₆, **SN76514**, where the harmonics are mixed with the signal from the first heterodyne oscillator Q₂₀₁. The output signal from the pre-mixer passes through the selective amplifier Q₁₀₇, Q₁₀₈ and Q₁₀₉, **2SC784**, which eliminates other signals except the 52.5 MHz second heterodyne signal.

A part of the output from the selective amplifier is rectified by the detectors D₁ and D₂, **1S1555**, and the DC output voltage is amplified by the DC amplifier Q₁₁₀, **2SC372**, and then fed to the LOCK lamp driver Q₁₁₁, **2SC372**, which turns the LOCK lamp on when the synthesizer is unlocked.

The output signal from the first IF amplifier Q₁₀₄ is fed to the second mixer Q₁₀₅, **2SK 19**, where the

incoming signal is mixed with the 52.5 MHz signal from the selective amplifier. The output of the second mixer becomes second IF signal of 2.0 through 3.0 MHz. The 2.0 to 3.0 MHz IF signal is then amplified by the second IF amplifier Q₄₀₁, **3SK40**, and then fed to the third mixer Q₄₀₂, **2SK19**. The third mixer converts the second IF signal into 455 kHz third IF signal. The VFO (maintuning) signal, which varies between 2,455 kHz and 3,455 kHz, is generated by the variable frequency oscillator Q₄₀₃, **2SC372**, and supplied to the third mixer through the buffer amplifier Q₄₀₄, **2SK19**. The 455 kHz IF signal from the third mixer is fed to the ceramic filter which is tuned to 455 kHz and has ± 3 kHz passband response to eliminate interference.

The signal is then amplified by the third amplifier Q₄₀₅ and Q₄₀₆, **2SC372**, and fed to the appropriate detector. The AM signal is detected by balanced diode detector D₄₀₂, **1N60AM**.

The balanced demodulator D₄₀₃ through D₄₀₆, **1N60AM**, is used for the detection of SSB and CW signals. The carrier signal for SSB and the beat frequency signal for CW which is generated by the BFO oscillator Q₄₀₈, **2SK19**, are fed to the balanced demodulator through buffer amplifier, Q₄₀₉, **2SC372**. The MODE switch shifts the BFO frequency 3 kHz lower than LSB position for USB and CW signal reception.

A part of the output from the last IF amplifier Q₄₀₆ is fed to the AGC (Automatic Gain Control) rectifier D₄₀₁, **1N60**. The rectified AGC voltage is then amplified by the AGC amplifier Q₄₀₇, **2SC372**, and fed to the Q₁₀₁, Q₄₀₁ and Q₄₀₅ to control the gain of these stages automatically when the incoming signal strength is varied. Thus the receiver audio output is not effected by the variation of the input signal strength which may be caused by phasing. The S-meter is placed in the emitter circuit of Q₄₀₇ in which the emitter current changes in accordance with the incoming signal strength.

The detected audio output is fed through the MODE switch and the VOLUME control potentiometer VR₁ to the audio amplifier integrated circuit

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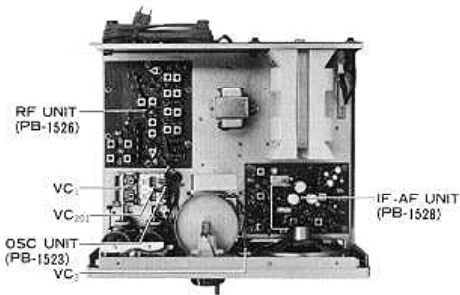
CIRCUIT DESCRIPTION (continued)

Q₄₁₀, **AN-214**, which utilizes OTL (Output Transformer Less) circuit delivering 3 watts to the speaker.

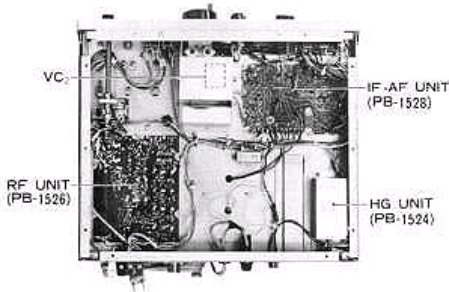
The power supply is designed to operate from either 100/110/117/200/220/234 volt AC 50/60 Hz or 12 volt DC (negative ground). For AC operation, +13.5 volts are supplied from full wave rectifier D₄₀₈ and D₄₀₉, **VO6B**.

The 13.5 volts are used for audio amplifier stage.

The DC voltage in both AC or DC operation is supplied to the voltage regulator Q₁₁₁, **2SD313**, to obtain an extremely stable 10 volt DC supply which is used by the various circuits. The 10 volt DC is further regulated by zener diode D₄₁₃, **BZ-091**, at 9 volts, and then supplied to the oscillators and harmonic generator circuits. When the AC supply fails, the DC voltage may be automatically supplied to the circuit through the diode D₄₁₀, **VO6B**, which prevents the rectified DC voltage from flowing into the battery.



TOP VIEW



BOTTOM VIEW

FREQUENCY f	1ST OSC fo ₁	1ST IF (fo ₁ -fi ₁) fi ₁	REF FREQ (1MHz x n) fh	2ND OSC (fo ₁ -fh) fo ₂	2ND IF (fi ₁ -fo ₂) fi ₂	3RD OSC fo ₂	3RD IF (fo ₃ -fi ₂) fi ₃
500kHz	55.5MHz	55.0MHz	3MHz	52.2MHz	2,500kHz	2,995kHz	455kHz
1,500	56.5	55.0	4	"	2,500	2,995	"
2,500	57.5	55.0	5	"	2,500	2,995	"
3,500	58.5	55.0	6	"	2,500	2,995	"
4,500	59.5	55.0	7	"	2,500	2,995	"
5,500	60.5	55.0	8	"	2,500	2,995	"
6,500	61.5	55.0	9	"	2,500	2,995	"
7,500	62.5	55.0	10	"	2,500	2,995	"
8,500	63.5	55.0	11	"	2,500	2,995	"
9,500	64.5	55.0	12	"	2,500	3,455	"
10,000	65.5	55.5	13	"	3,000	3,455	"
11,000	66.5	55.5	14	"	3,000	3,455	"
12,000	67.5	55.5	15	"	3,000	3,455	"
13,000	68.5	55.5	16	"	3,000	3,455	"
14,000	69.5	55.5	17	"	3,000	3,455	"
15,000	70.5	55.5	18	"	3,000	3,455	"
16,000	71.5	55.5	19	"	3,000	3,455	"
17,000	72.5	55.5	20	"	3,000	3,455	"
18,000	73.5	55.5	21	"	3,000	3,455	"
19,000	74.5	55.5	22	"	3,000	3,455	"
20,000	75.5	55.5	23	"	3,000	3,455	"
21,100	76.5	55.4	24	"	2,900	3,355	"
22,200	77.5	55.3	25	"	2,800	3,255	"
23,300	78.5	55.2	26	"	2,700	3,155	"

24,400	79.5	55.1	27	"	2,600	3,055	"
25,500	80.5	55.0	28	"	2,500	2,955	"
26,600	81.5	54.9	29	"	2,400	2,855	"
27,700	82.5	54.8	30	"	2,300	2,755	"
28,800	83.5	54.7	31	"	2,200	2,655	"
29,900	84.5	54.6	32	"	2,100	2,555	"

Table 3 Frequency Relationship

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MAINTENANCE & ALIGNMENT

The FRG-7 has been carefully aligned and tested at the factory using the precise test instruments before shipment and, with normal usage, it should not require other than the usual attention given to any electronic equipment. Service or replacement of major component may require substantial realignment, however, under no circumstances, should realignment be attempted unless the operation of the receiver is fully understood and the malfunction has been fully analyzed and traced to misalignment. Service work should only be performed by experienced personnel using proper test equipment.

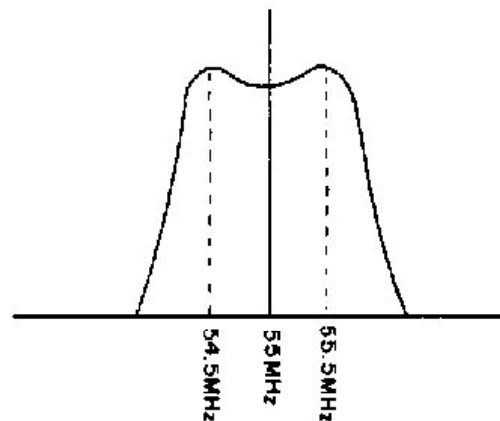


Fig. 5

TEST EQUIPMENT REQUIRED

(1) RF Signal Generator; Hewlett-Packard Model 606A or equivalent with one volt output at an impedance of 50 ohms and a frequency coverage to 30 MHz.

(2) Vacuum Tube Volt-Ohm Meter (VTVM): Hewlett-Packard Model 401B or equivalent VTVM with RE probe workable to 60 MHz.

(3) Sweep Generator and Oscilloscope workable to 60 MHz.

(4) Frequency Counter: Yaesu YC-355D or equivalent workable to 60 MHz.

(2) Balanced Mixer, VR₁₀₁, TC₁₀₆

Set the BAND switch to A and the MHz dial to 0. Disconnect the antenna, and connect its output to antenna terminal. Tune the receiver to the internal spurious signal at 910 kHz. Adjust VR₁₀₁ and TC₁₀₅ for minimum S-meter indication.

(3) Antenna Coil and Trimmer, T₁₀₁ ~ T₁₀₄

Connect the signal generator output to the antenna terminal SW₂ and connect SW₁ and BC terminals with a copper wire.

Set the signal generator to 0.5 MHz, the BAND to A and PRESELECT to 0.5. Tune the receiver to the signal generator signal. Adjust T₁₀₁ for maximum S-meter reading. Repeat this procedure at the frequencies shown in Table 4.

RF UNIT PB-1526

(1) 55 MHz Passband Circuit, T₁₀₅ ~ T₁₀₈

Set the BAND switch to D and the MHz dial to 20 MHz position. Disconnect the antenna. Connect the sweep generator output between TP₁₀₃ and TP₁₀₂ (ground), and the oscilloscope input between TP₁₀₄ and TP₁₀₅ (ground).

Set the center frequency of the sweep generator to 55 MHz and align T₁₀₅ through T₁₀₈ until the scope indicates the curve shown in Fig. 5. Disconnect the sweep generator and the scope.

FREQ	BAND	PRESELECT	ALIGNMENT
0.5MHz	A	0.5	T101
1.6MHz	A	1.6	TC101
1.6MHz	B	1.6	T102
4.0MHz	B	4.0	TC102
4.0MHz	C	4.0	T103
11.0MHz	C	11.0	TC103
11.0MHz	D	11.0	T104
30.0MHz	D	30.0	TC104

Table 4

MAINTENANCE & ALIGNMENT (continued)

Disconnect the signal generator and the copper wire between SW₁ and BC.

IF AF UNIT PB-1528

(4) 52.5 MHz Selective Filter, T₁₀₉ ~ T₁₁₆

Disconnect the input from the oscillator unit at TP₁₀₁. Connect the sweep oscillator output between TP₁₀₇ and ground, and the scope between TP₁₀₉ and ground. Set the center frequency of the sweep generator to 52.5 MHz. Adjust T₁₀₉ to T₁₁₆ until the scope screen shows the curve shown in Fig. 6.

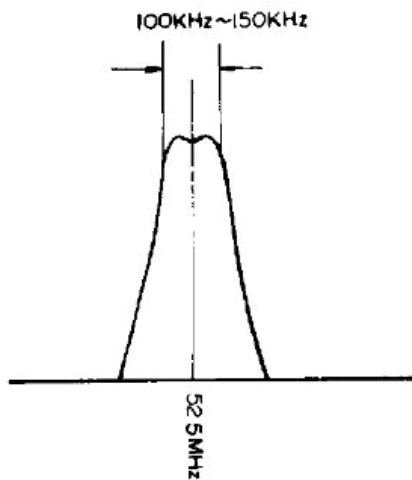


Fig. 6

Disconnect the sweep generator and scope and reconnect the wiring at TP₁₀₁. After completion of the above procedures make sure that the RF voltage between TP₁₁₀ and ground is approximately 0.3 to 0.5 volt RMS. If not, repeat procedure.

(5) LOCK Level, VR₁₀₂

Adjust VR₁₀₂ until the LOCK lamp turns off at any MHz setting of the MHz dial.

OSC UNIT PB-1523

(1) MHz Setting, T₂₀₁ ~ TC₂₀₁

(1) Main Tuning Dial, T₄₀₃, TC₄₀₃

The following alignment should be done after warm-up of the receiver.

Set the dial hair line to the center of the dial window and FINE TUNING control to 12 o'clock position. When the main tuning dial is rotated until it stops over 1000 scale, ▲ mark should be within 5 m/m from the hair line.

Set the MODE switch to LSB and Mhz dial to 0. Set the main tuning dial to 1000, then beat tone will be heard. Adjust T₄₀₃ for zero beat. Set the main tuning dial to 0 and adjust TC₄₀₃ for zero beat. Repeat above procedure until the tracking is completed.

(2) 2nd IF Tracking, TC₄₀₁, TC₄₀₂, T₄₀₁, T₄₀₂

Connect the signal generator to the antenna terminal SW₂ and set its frequency to 7.1 MHz. Tune the receiver to the signal from the signal generator. Set the output voltage from the signal generator for S-3 reading on S-meter. Adjust TC₄₀₁ and TC₄₀₂ for maximum S-meter reading. Set the signal generator to 7.9 MHz and tune the receiver to 7.9 MHz signal. Adjust TC₄₀₁ and T₄₀₂ for maximum S-meter reading. Repeat these procedures until the tracking is completed.

(3) 3rd IF, T₄₀₄, TC₄₀₅

Set the signal generator to 7.5 MHz and tune the receiver to this frequency. Adjust T₄₀₄ and T₄₀₅ for maximum S-meter reading. Adjust signal level so as not to saturate.

(4) S-meter Sensitivity, VR₄₀₁

Set the output level of the signal generator to 100 dB. And tune the receiver for maximum S-meter reading. Adjust VR₄₀₁ for S-meter full scale.

Connect the signal generator to antenna connector SW₂ and set its frequency to 3.5 MHz. Tune the receiver to the signal generator signal. Adjust T₂₀₂ carefully until the LOCK lamp turns off at the center of 3 MHz scale on the MHz dial. Set the signal generator frequency to 27.5 MHz and tune the receiver to this signal. Carefully adjust TC₂₀₁ until the LOCK lamp turns off at the center of 27 MHz scale on the MHz dial. Repeat above procedures until the LOCK lamp turns off at the center of every MHz scale, from 0 to 29 MHz. Disconnect the signal generator.

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Disconnect the signal generator.

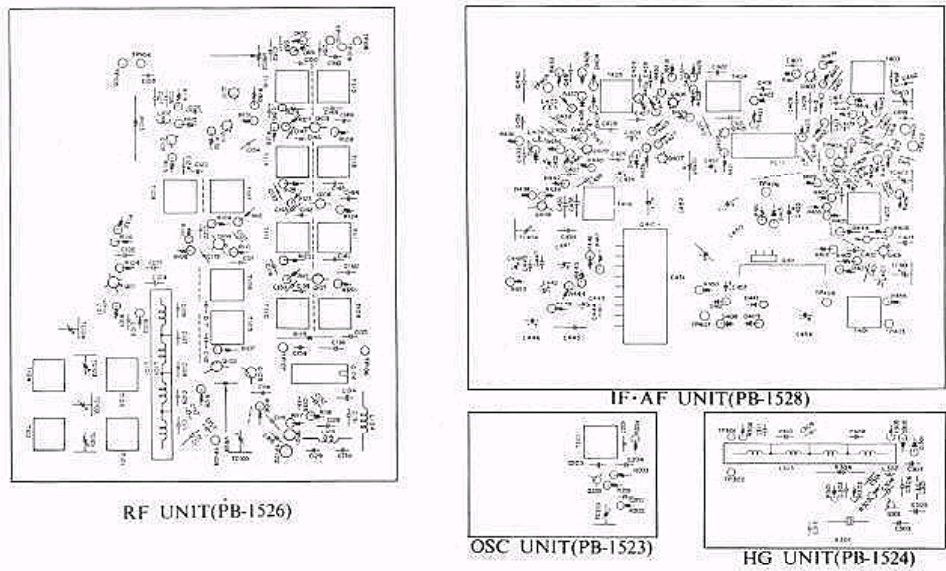
(5) BFO Frequency, T₄₀₆, TC₄₀₄

Connect a frequency counter to TP₄₀₅. Set the MODE switch to LSB. Adjust T₄₀₆ for 457 kHz on the frequency counter reading. Set the MODE switch to USB/CW and adjust TC₄₀₄ for 453 kHz.

Voltage Chart

	E (S)		C (D)		B (G)				E (S)		C (D)		B (G)			
Q101	1.5		4.2		G ₁ 1.5 G ₂ 4.0				Q301	0.2		8.0		-1.1		
Q102	1.6		9.0		0				Q401	2.0		9.0		G ₁ 1.6 G ₂ 2.7		
Q103	2.2		9.0		0				Q402	1.8		9.2		0		
Q104	0.5		9.0		G ₁ 0 G ₂ 4.5				Q403	1.8		3.5		2.1		
Q105	2.0		9.2		0				Q404	0.5		7.8		0		
Q106	0.7		9.2		1.3				Q405	4.3		8.5		5.0		
Q107	1.1		9.2		1.7				Q406	1.4		9.1		2.0		
Q108	1.4		8.8		2.0				Q407	0.01		8.7		0.3		
Q109	0		0.02		0.5				Q408	1.4		6.8		0		
Q110	0		9.5		0.02				Q409	2.2		7.0		3.3		
Q201	1.8		7.7		1.2				Q411	9.5		13.5		10.0		
	1	2	3	4	5	6	7	8	9	10	11	12	13	14		
Q106	0	8.2	8.2	4.2	2.6	0	0	0	2.2	4.1	4.1	4.2	7.3	0		
Q410	6.5	0	7.8	11.0	6.5	0	6.5	12	13.5	-	-	-	-	-		

BAND 4.0 ~ 11.0 MHz 7 MODE USB/CW Measured with VTVM Values are in VOLTS DC



Part List

MAIN CHASIS			5		MJ-164
PB Printed Circuit Board			6 (P-6)		#4003A
1390 (A ~ Z)	Lamp Board		7		SG-8050-07
1525 (A ~ Z)	Switch Board				
1569 (A ~ Z)	LED Board				
			F Fuse		
			1		0.5 A
D Light Emitting Diode					
1	SL-103				
			FH Fuse Holder		
			1	SN01301	
R Resistor					
CARBON FILM					
1, 2	1/4 W	10 Ohm	RF UNIT		
5, 9	1/4 W	68 Ohm	PB Printed Circuit Board		
3, 6, 12	1/4 W	100 Ohm	1526 (A ~ Z)		
4	1/4 W	1 KOhm			
7, 11	1/4 W	3.3 KOhm	Q IC, FET & Transistor		
10	1/4 W	10 KOhm	106	IC	SN76514N
8	1/4 W	22 KOhm	101, 104	FET	3SK40M
CARBON COMPOSITION			102, 103, 105	FET	2SK19GR
13	1/2 W	56 Ohm	110, 111	Tr	2SC372Y
14	1/2 W	68 Ohm	107 ~ 109	Tr	2SC784R (O)
			D Diode		
VR Potentiometer			101, 102	Si	1S1555
1	EVH-BOAS 20A14	10 KA			
			PL Pilot Lamp		
			1, 2, 3, 4, 5, 8		BQ041-32404A
C Capacitor			6, 7		BQ154-33811A
CERAMIC DISC					
4	50 WV	22 PF (SL)			
3	50 WV	33 PF (SL)	R Resistor		
MYLAR			CARBON FILM		
2	50 WV	0.03 mF	132	1/4 W	100 Ohm
6, 7	50 WV	0.047 mF	102, 103, 107, 111, 112	1/4 W	220 Ohm
1	50 WV	0.068 mF	115, 125, 129		
			119, 122	1/4 W	330 Ohm
			118, 128	1/4 W	470 Ohm
			104, 105	1/4 W	560 Ohm
VC Variable Capacitor			117, 120, 124	1/4 W	1 KOhm
1	C123A119	300 PF x 2	114	1/4 W	3.3 KOhm
2	C134ER20	320 PF x 2	126	1/4 W	4.7 KOhm
3	TSN 150S x 05	5 PF	136, 137	1/4 W	6.8 KOhm
			130	1/4 W	8.2 KOhm
PT Transformer			127	1/4 W	15 KOhm
1	52-51 (51-50)		123	1/4 W	22 KOhm
			121	1/4 W	27 KOhm

			109, 110	¼ W	33 KOhm
CT Choke			116	¼ W	39 KOhm
1	50-51		133	¼ W	47 KOhm
			101, 106, 108, 113, 131, 135	¼ W	100 KOhm
			134	¼ W	330 KOhm
M Meter			123	¼ W	390 KOhm
1	KM-005		CARBON COMPOSITION		
			135	1/8 W	100 KOhm
			134	1/8 W	330 KOhm
SP Speaker			VR Potentiometer		
1	SA-128		102	EVL-S3A-B13	1 KB
			101	EVL-S3A-B53	5 KB
S Switch			C Capacitor		
1	ATT	ESL-3037	CERAMIC		
4	TONE	ESL-3037	159	50 WV	1 PF (CH)
2	BAND	ESR-E264R20	117, 138, 142, 146	50 WV	2 PF (CH)
3	MODE	ESR-E264R20	106, 109, 122, 160	50 WV	3 PF (CH)
5	POWER	8H2011	111, 130, 150	50 WV	5 PF (CH)
6	LAMP	8H2011	128, 131	50 WV	10 PF (CH)
			119	50 WV	15 PF (CH)
			129, 157, 158, 161	50 WV	22PF (CH)
J Connector			110	50 WV	27 PF (CH)
1		JSO-239	107, 108	50 WV	33 PF (CH)
2		SQ-2450-03	139, 143, 168	50 WV	47 PF (CH)
3		SG-8050-07	112414	50 WV	83 PF (CH)
4		SG-8481	114	50 WV	100 PF (SL)

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Part List (continued)

147, 166, 167	50 WV	0.001 MF	301, 302	Ge	1N60FM
101 ~ 105, 112, 113 115, 116, 118, 120, 121, 123 125, 127, 135 ~ 137, 140 141, 144, 145, 148, 149 151, 156, 162 ~ 165	50 WV	0.01 MF			
			X Crystal		
			301	HC-5/U	1MHz
126, 132 ~ 134	50 WV	0.047 MF			
			R Resistor		
			305, 306	¼ W	100 Ohm
TC Trimmer Capacitor			303, 304	¼ W	220 Ohm
101	ECV-1ZW 20x32	20 PF	302	¼ W	10 KOhm
102 ~ 104	ECV-1ZW 40x32	40 PF	301	¼ W	100 KOhm
L Inductor			C Capacitor		
102	RFC	68 MH		DIPPED MICA	
101	LPF	#220051	302	50 WV	180 PF
103		#220053		CERAMIC	
104		#220054	310	50 WV	8 PF (CH)
			311	50 WV	15 PF (CH)
			312	50 WV	22 PF (CH)
T Transformer			306, 309	50 WV	27 PF (CH)
101		#220046	308	50 WV	33 PF (CH)
102		#220047	305	50 WV	68 PF (CH)
103		#220048	301, 303, 304, 307	50 WV	0.01 MF
104		#220049			
105 ~ 106		#220050			
			L Inductor		
			302	RFC	4.7 mH
OSC UNIT			301	RFC	1 mH
PB Printed Circuit Board			303	LPF	#220051
1523 (A ~ Z)					
				IF * AF UNIT	
Q Transistor			PB Printed Circuit Board		
201		2SC784R (O)	1528 (A ~ Z)		
R Resistor			Q IC, FET & Transistor		
	CARBON FILM		410	IC	AN-214
204	¼ W	100 Ohm	401	FET	3SK40M
203	¼ W	1 KOhm	402, 404, 408		2SK19GR
202	¼ W	4.7 KOhm	403, 405 ~ 407, 409		2SC372Y
201	¼ W	22 KOhm	411		2SD313
C Capacitor			D Diode		
	CERAMIC		401 ~ 407	Ge	1N60FM

203	50 WV	22PF (CH)	408 ~ 410	Si	V06B
204	50 WV	33 PF (CH)	413	Zener	BZ091
201, 202	50 WV	0.01 MF	412	Zener	WZ110
VC Variable Capacitor			FL Ceramic Filter		
201	C521	30PFx2	401	455 KHz	LFC-6
TC Trimmer Capacitor			R Resistor		
201	ECV-1ZW 10x32	10 PF	CARBON FILM		
			430	50 WV	15 Ohm
			451	50 WV	32 Ohm
T Transformer			406, 415, 423, 432, 442	50 WV	100 Ohm
201		#220052	444	50 WV	120 Ohm
			407, 419, 424, 433, 437	50 WV	220 Ohm
			441, 452		
HG Unit			450	50 WV	390 Ohm
PB Printed Circuit Board			431, 448	50 WV	470 Ohm
1523 (A ~ Z)			404, 436, 445	50 WV	560 Ohm
			411, 422, 438	50 WV	1 KOhm
			420	50 WV	1.5 KOhm
Q Transistor			413, 449, 453, 454, 457	50 WV	2.2 KOhm
301		2SC372Y	417, 418, 427, 429, 439	50 WV	3.3 KOhm
			427	50 WV	3.5 KOhm
			403, 428	50 WV	10 KOhm
D Diode			447	50 WV	15 KOhm

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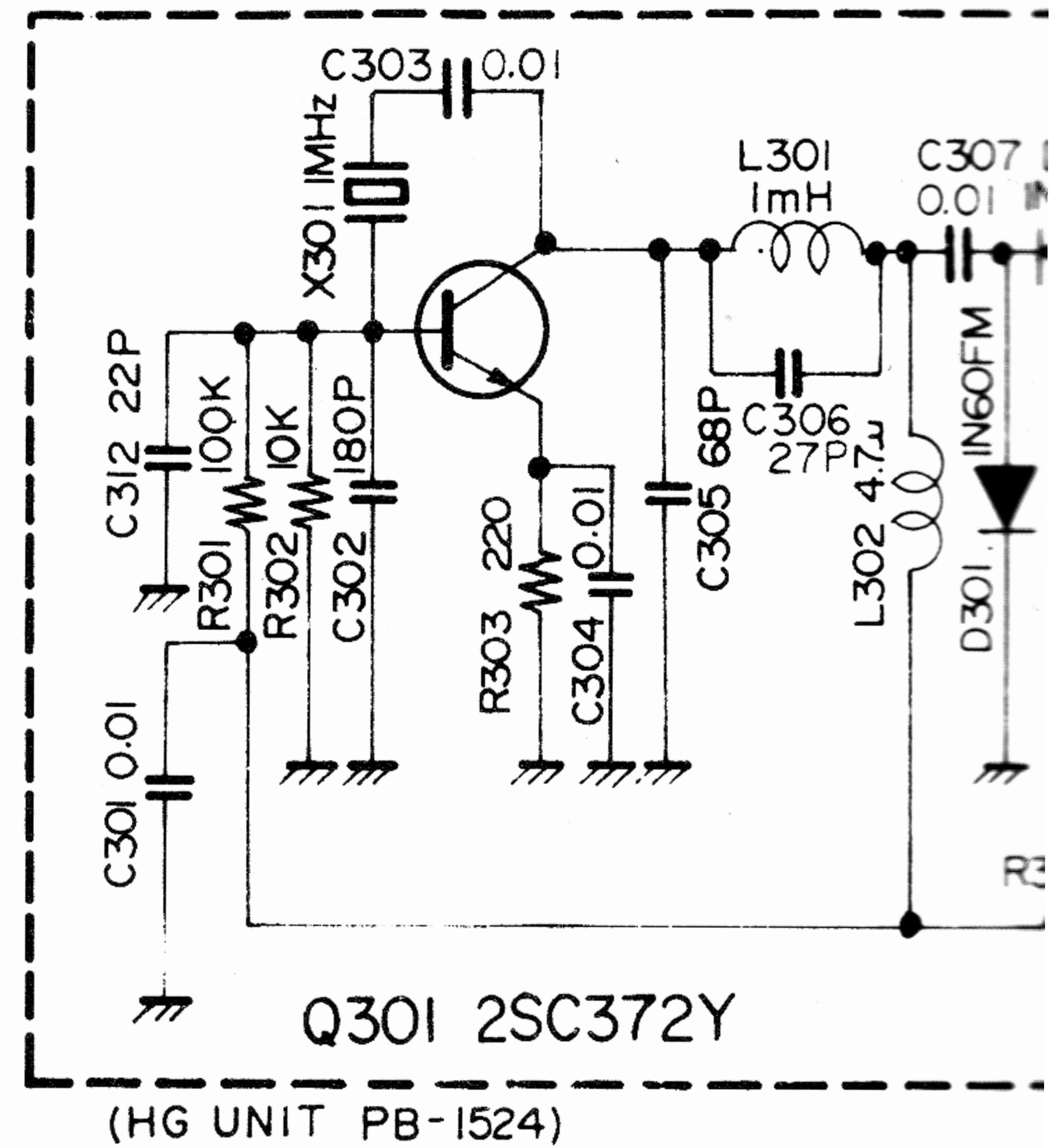
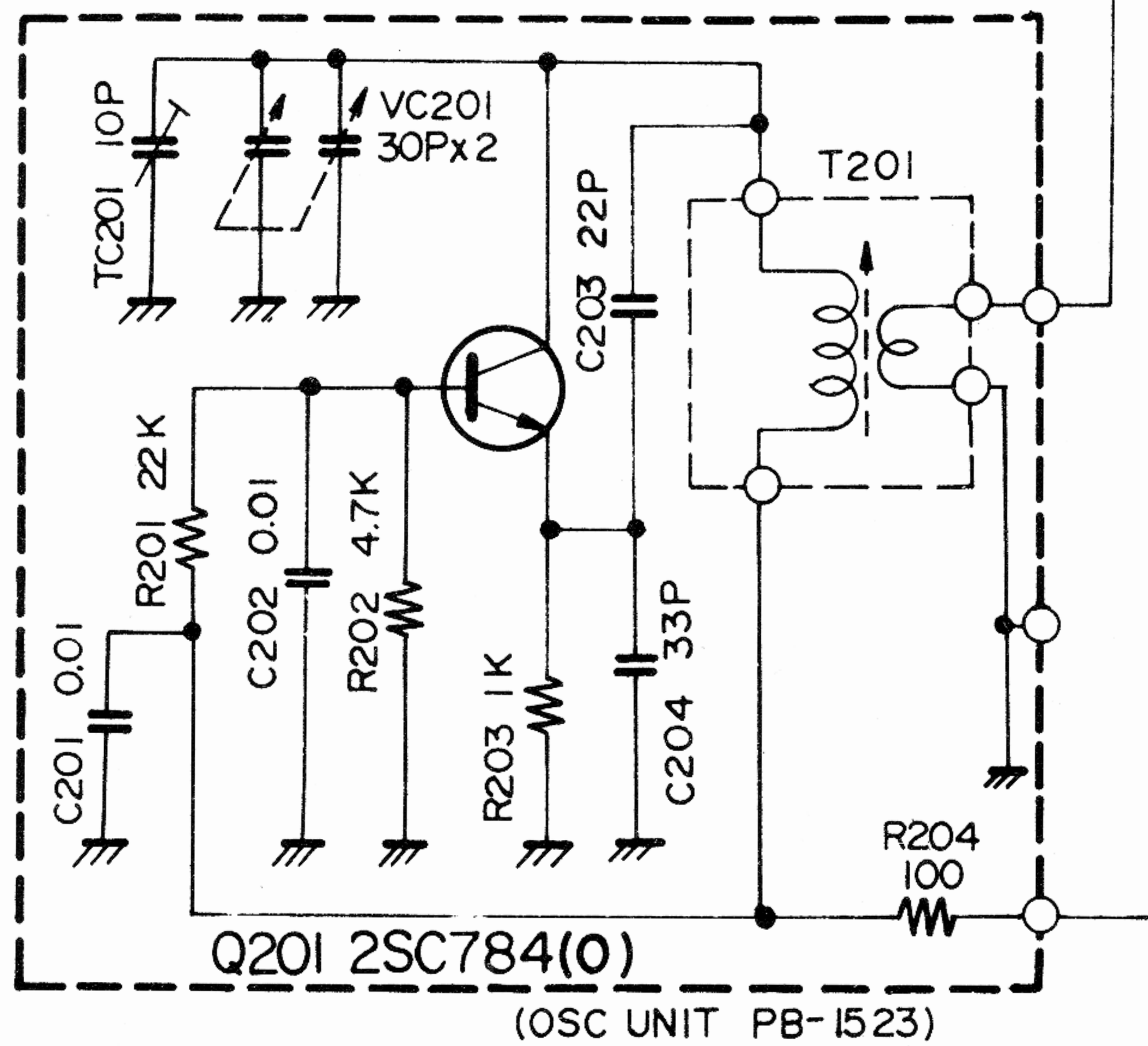
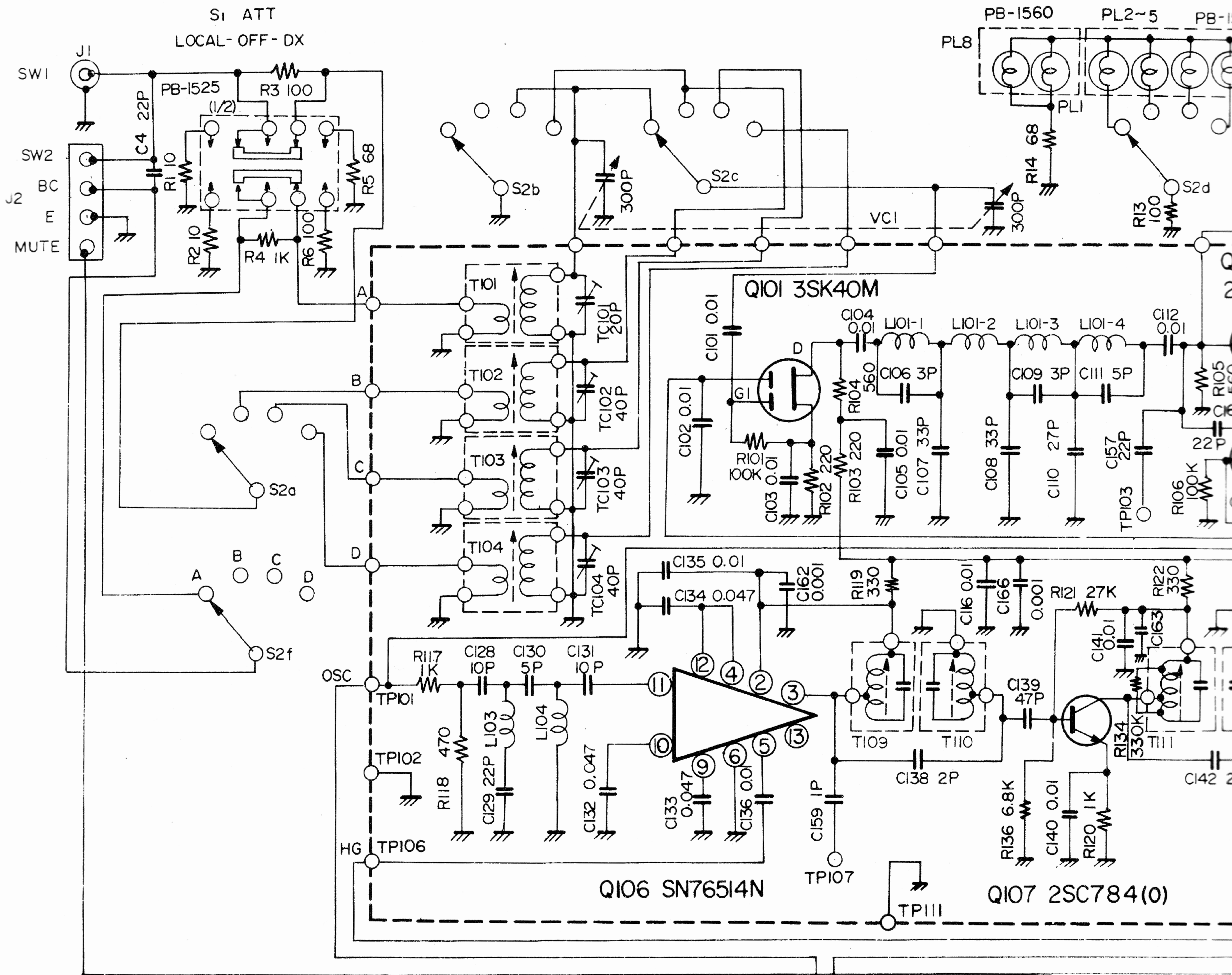
Part List (continued)

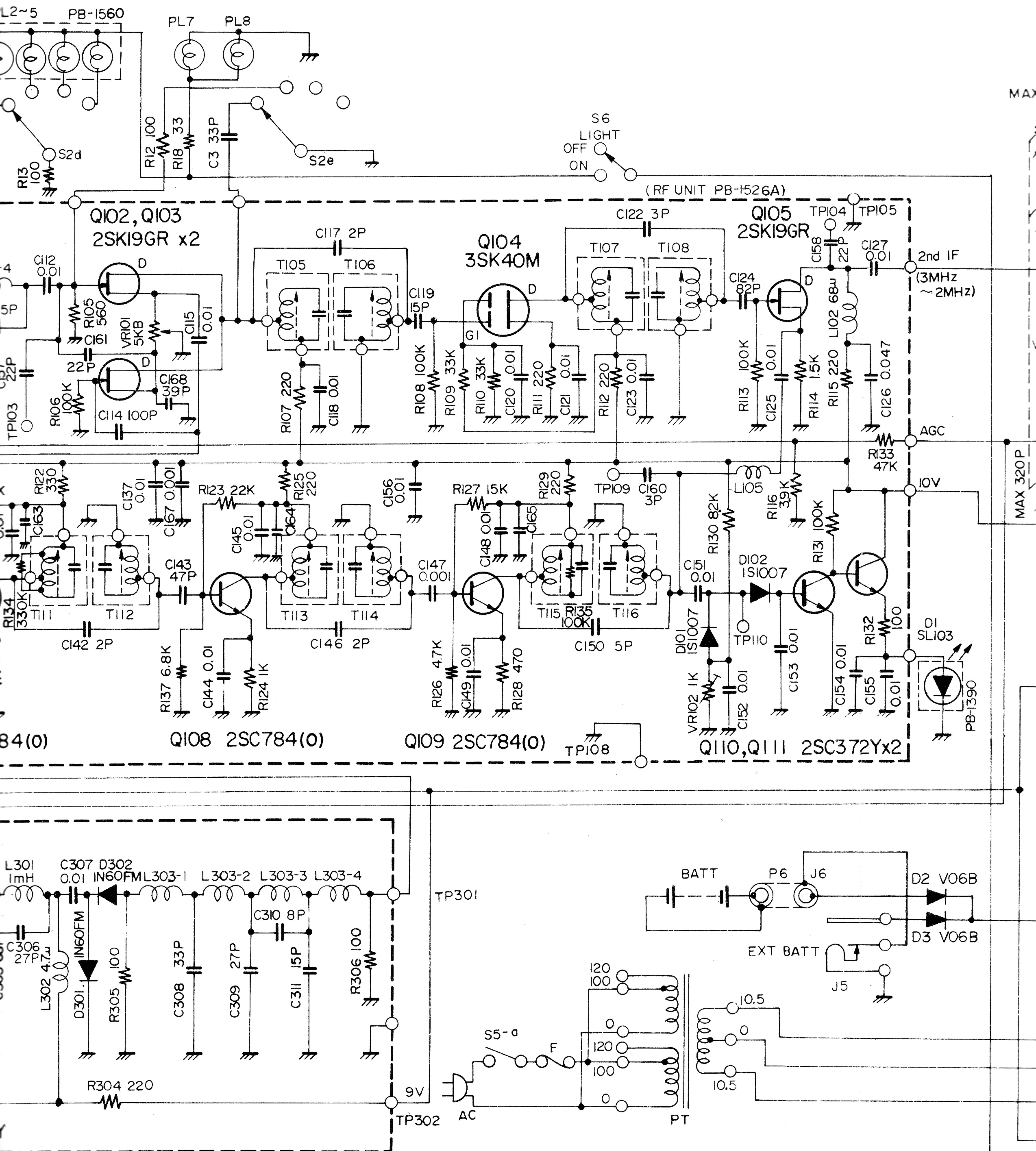
410, 425, 443	¼ W	18 KOhm
446	¼ W	22 KOhm
409	¼ W	33 KOhm
401, 445, 456	¼ W	47 KOhm
405, 440	¼ W	68 KOhm
402, 408, 414, 434	¼ W	100 KOhm
435	¼ W	150 KOhm
421	¼ W	220 KOhm
VR Potentiometer		
401	EVL-S0A-B32	300 Ohm B
C Capacitor		
DIPPED MICA		
462	50 WV	20 PF
458	50 WV	33 PF
434	50 WV	100 PF
410	50 WV	120 PF
435, 436	50 WV	620 PF
409	50 WV	680 PF
408	50 WV	1000 PF
CERAMIC		
420, 424	50 WV	1 PF (CH)
438	50 WV	10 PF (CH)
412	50 WV	22 PF (CH)
459	50 WV	100 PF (UJ)
427, 428	50 WV	100 PF (SL)
415	50 WV	150 PF (SL)
414	50 WV	220 PF (SL)
430	50 WV	0.007 MF
401, 402, 405 ~ 407 416, 417, 422, 425, 431 439, 461	50 WV	0.01 MF
403, 404, 413, 418, 419 421, 423, 429, 437, 461	50 WV	0.047 MF
MYLAR		
444	50 WV	0.003 MF
456	50 WV	0.0047 MF
433	50 WV	0.01 MF
447, 449	50 WV	0.03 MF
440, 445	50 WV	0.2 MF
ELECTROLYTIC		
448	16 WV	1 MF
432, 457	16 WV	2.2 MF
453, 455	16 WV	10 MF
442, 443	16 WV	33 MF

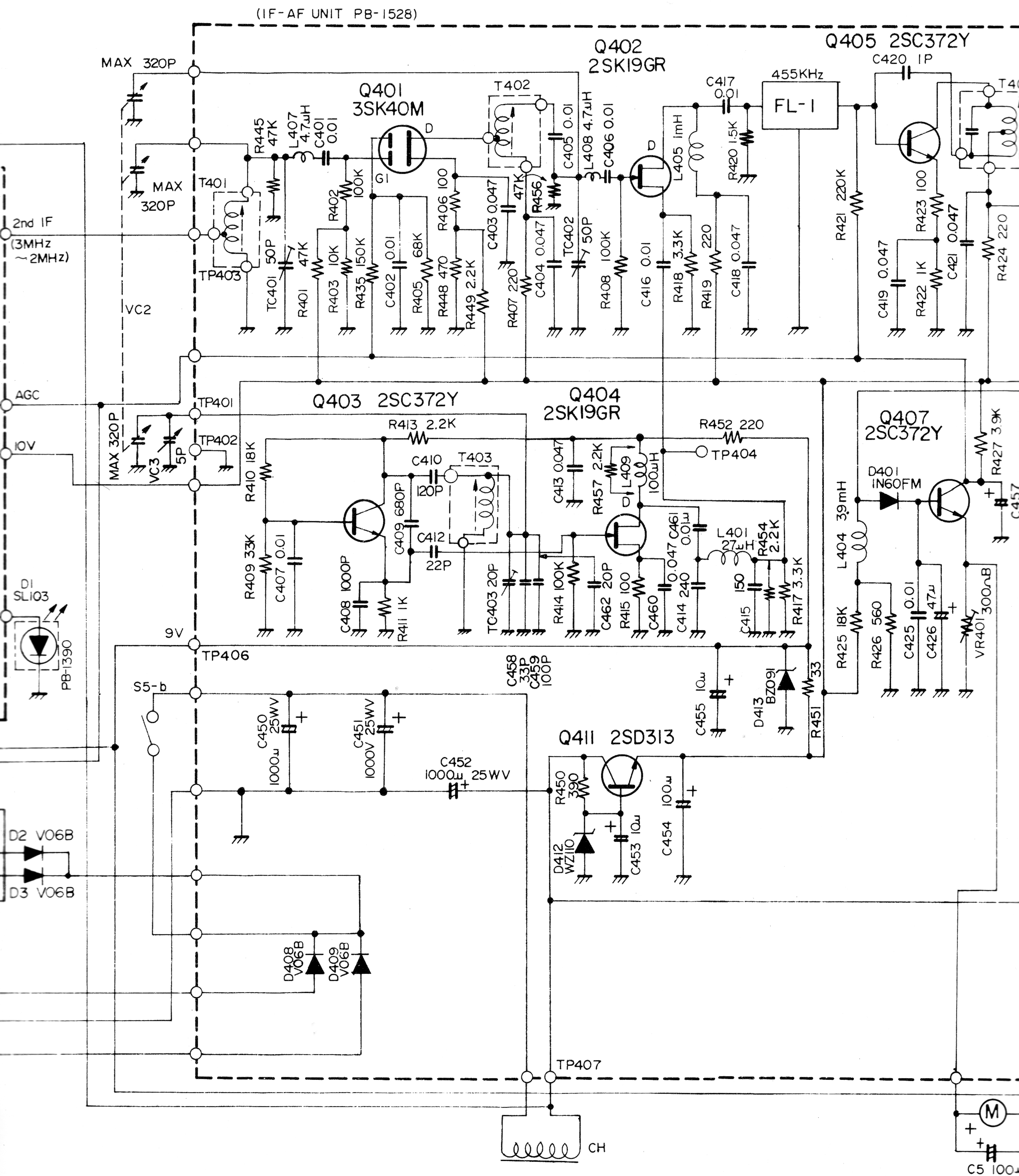
426	16 WV	47 MF
441, 446, 454, 463	16 WV	100 MF
450 ~ 452	16 WV	1000 MF
TC Trimmer Capacitor		
403, 404	ECV-1ZW 20x32	20 PF
401, 402	ECV-1ZW 50x32	50 PF
L Inductor		
407, 408	RFC	4.7 mH
401	RFC	27 mH
409	RFC	100 mH
405	RFC	1 mH
402 ~ 404, 406	RFC	3.9 mH
T Transformer		
401		#220060
402		#220061
403		#220062
404, 405		R12-4097
406		R12-4099

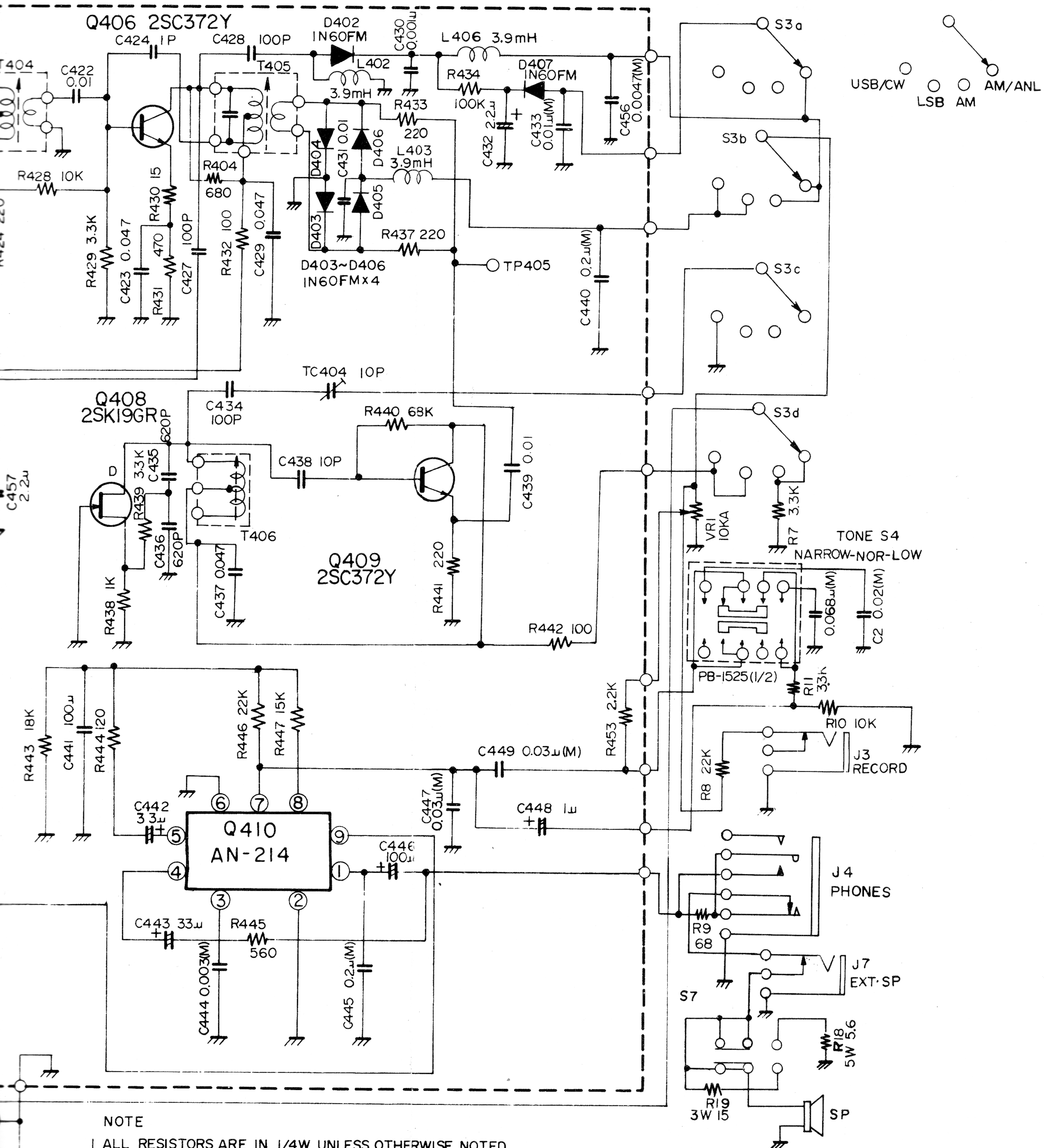
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NOTE

- 1 ALL RESISTORS ARE IN 1/4W UNLESS OTHERWISE NOTED.
- 2 ALL CAPACITORS ARE IN μ F UNLESS OTHERWISE NOTED.
- 3 ALL ELECTROLYTIC CAPACITORS ARE 16WV UNLESS OTHERWISE NOTED.

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CIRCUIT DIAGRAM